



Term Evaluation (Odd) Semester Examination September 2025

Roll no.....

Name of the Course and semester: MCA I semester & MCA (AI&DS) I semester

Name of the Paper: *Probability and Statistics*

Paper Code: TMC-111 / TMD-101

Time: 1.5 hour

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub questions
- (ii) Each question carries 10 marks.
- (iii) Please specify COs against each question.

Q1. (10 Marks)

a. What is statistics? Define primary and secondary data in statistics. CO1

OR

b. In a bolt factory, machines A,B and C produce 25, 35 and 40% of the total output respectively. Of their outputs 5, 4 and 2% respectively are defective bolts. If a bolt is chosen at random from the combined output, what is the probability that it is defective? If a bolt chosen at random is found to be defective, what is the probability that it was produced by B? CO1

Q2. (10 Marks)

a. Let X be a random variable of sum of two numbers in throwing two fair dice. Find the probability distribution of X, mean and variance. CO1

OR

b. A random variable X has the following probability function values CO1

x	0	1	2	3	4	5	6
$P(x = X)$	0	k	$2k$	$3k$	$2k$	k^2	$2k^2$

- (i) Find the value of k
- (ii) Determine the distribution function of random variable X
- (iii) Evaluate $P(X < 5)$, $P(X \geq 5)$, $P(1 < X < 6)$

Q3. (10 Marks)

a. Calculate the first four moments about the mean for the following distribution CO1

x	6	7	8	9	10	11	12
f	3	6	9	13	8	5	4

OR

b. Define the term Mathematical Expectation, Also derive the following properties

- (i) $E(X + Y) = E(x) + E(y)$
- (ii) $E(XY) = E(X) \cdot E(Y)$

CO1



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- Q4. (10 Marks)
- a. If the sum and product of the mean and variance of a binomial distribution are 24 and 128 . Find the distribution? CO2
- OR
- b. If X is a poisson variate such that $P[X=1] = P[X=2]$. Find CO2
(i) $P[X=0]$ (ii) Mean (iii) Variance
- Q5. (10 Marks)
- c. Daily income of workers follows the Normal distribution with mean Rs. 1000 and standard deviation Rs.100. Find the probability of income CO2
(i) Less than 1100 Rs.
(ii) More than 1100 Rs.
(iii) Less than 790 Rs.
Given that $P(Z=1) = 0.3413$, $P(Z=2.1) = 0.4821$
- OR
- a. If X is a continuous random variable with probability density function give by
- $$f(x) = \begin{cases} kx & 0 \leq x \leq 2 \\ 2k & 2 < x < 4 \\ -kx + 6k & 4 \leq x \leq 6 \end{cases} \quad \text{CO2}$$
- Find (i) Value of K (ii) Mean of X